

Computer Mathematics, AI and Functional Programming

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1950's – The Logic Theorist and the birth of AI

- Proved theorems from *Principia Mathematica*.
- Reasoning as search – *heuristics*.
- Invented a *programming language for list processing* (predecessor to LISP).

THE LOGIC THEORY MACHINE
A COMPLEX INFORMATION PROCESSING SYSTEM

by

Allen Newell and Herbert A. Simon

P-868

June 15, 1956

“But, although the **program of LT does not change** as it accumulates experience in solving problems, **learning does take place** in one very important respect. The program stores the new theorems it proves, and **these theorems are then available as building blocks for proofs of subsequent theorems.**”

1960's: Optimism!

*“Within ten years a digital computer will **discover and prove an important new mathematical theorem**”*

(Newell & Simon 1958)

- Important algorithms of the 60's:
 - **DPLL** for SAT-solving (Davis-Putnam-Logemann-Loveland).
 - **Resolution** for first-order logic (Robinson)
 - Led to development of **PROLOG-language**

Axioms, conjecture

Automated Prover
Reasoning-as-search

Proved? Y/N

1970's: Boom and bust

Disappointment - what had been promised not achieved.

- No discovery of important new theorems,
- No machine translation,
- No reliable image recognition....

Limitations in computer mathematics:

- Automatic and slow. No proofs, or
- Proof checkers – little or no automation.



1970's: Boom and bust

A step back from full automation of proofs:

Interactive theorem proving - best of both worlds?

Edinburgh LCF (Milner et al)

- Program small *tactics* to automate (parts of) proofs.
- Check proof steps - small trusted core.
- Build up *theories* - theorems and proofs.
- Needed powerful *meta-language*: **ML**



1980-90's

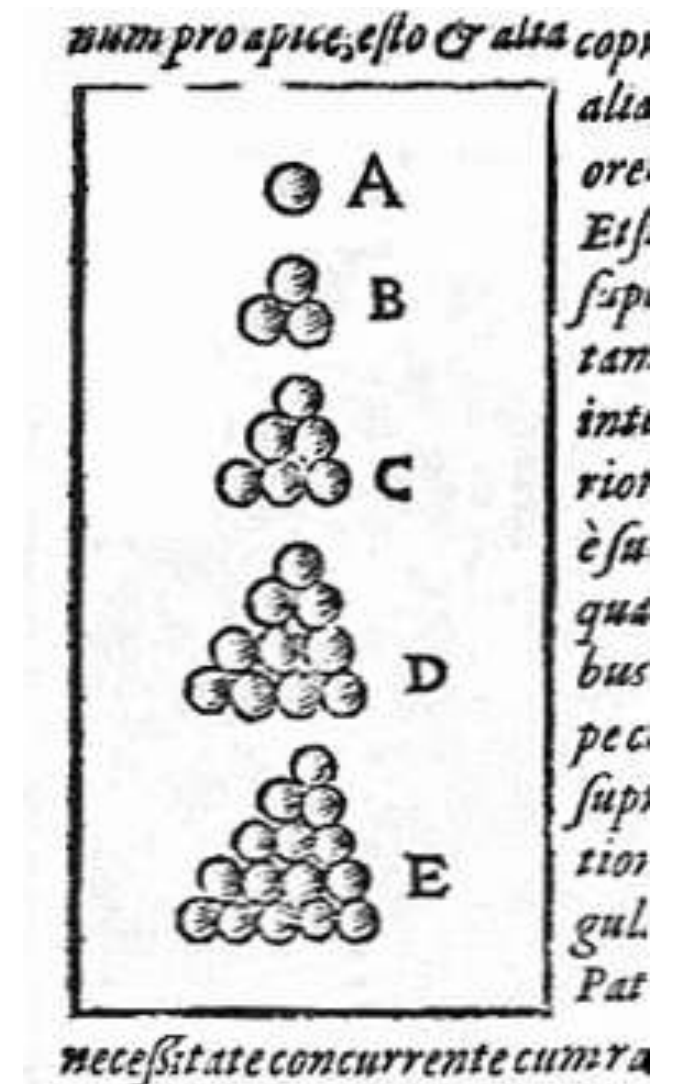
- “Meta-language” soon became a programming language in its own right: **Standard ML**.
- Many decedents of LCF: e.g. Isabelle/HOL, HOL (Light), Coq.



1990's-00's

Increasing amount of mathematics
formalised in interactive theorem proving
libraries:

- **Newtons Principia** (Fleuriot 1998)
 - Small error found.
- **Kepler's Conjecture** (Hales 1998)
 - Flyspeck project (2003 – 2014)



Today

~~“Within ten years a digital computer will discover and prove an important new mathematical theorem”~~

- Faster computers - come-back of brute force reasoning-as-search.
- **Boolean Pythagorean Triples Problem:** proved automatically SAT-solver (Heule & Kullmann 2016)
 - 200 TB of proof...
- **Hammers** - interactive provers connect to automated.
 - Machine learning selects (likely) relevant facts.
 - External automated prover.
 - Proofs recreated automatically in trusted setting.
- **Challenge:** Can computers discover interesting conjectures themselves?

Demo: automate my homework

```
fun sorted :: "nat list  $\Rightarrow$  bool"
  where "sorted [] = True"
    | "sorted [x] = True"
    | "sorted (x1#x2#xs) = ((x1  $\leq$  x2)  $\wedge$  sorted (x2#xs))"

fun ins :: "nat  $\Rightarrow$  nat list  $\Rightarrow$  nat list"
  where "ins x [] = [x]"
    | "ins x (y#ys) = (if (x  $\leq$  y) then (x#y#ys) else y#(ins x ys))"

fun isort :: "nat list  $\Rightarrow$  nat list"
  where "isort [] = []"
    | "isort (x#xs) = ins x (isort xs)"

theorem my_homework: "sorted (isort x)"
```

Hipster and bugs

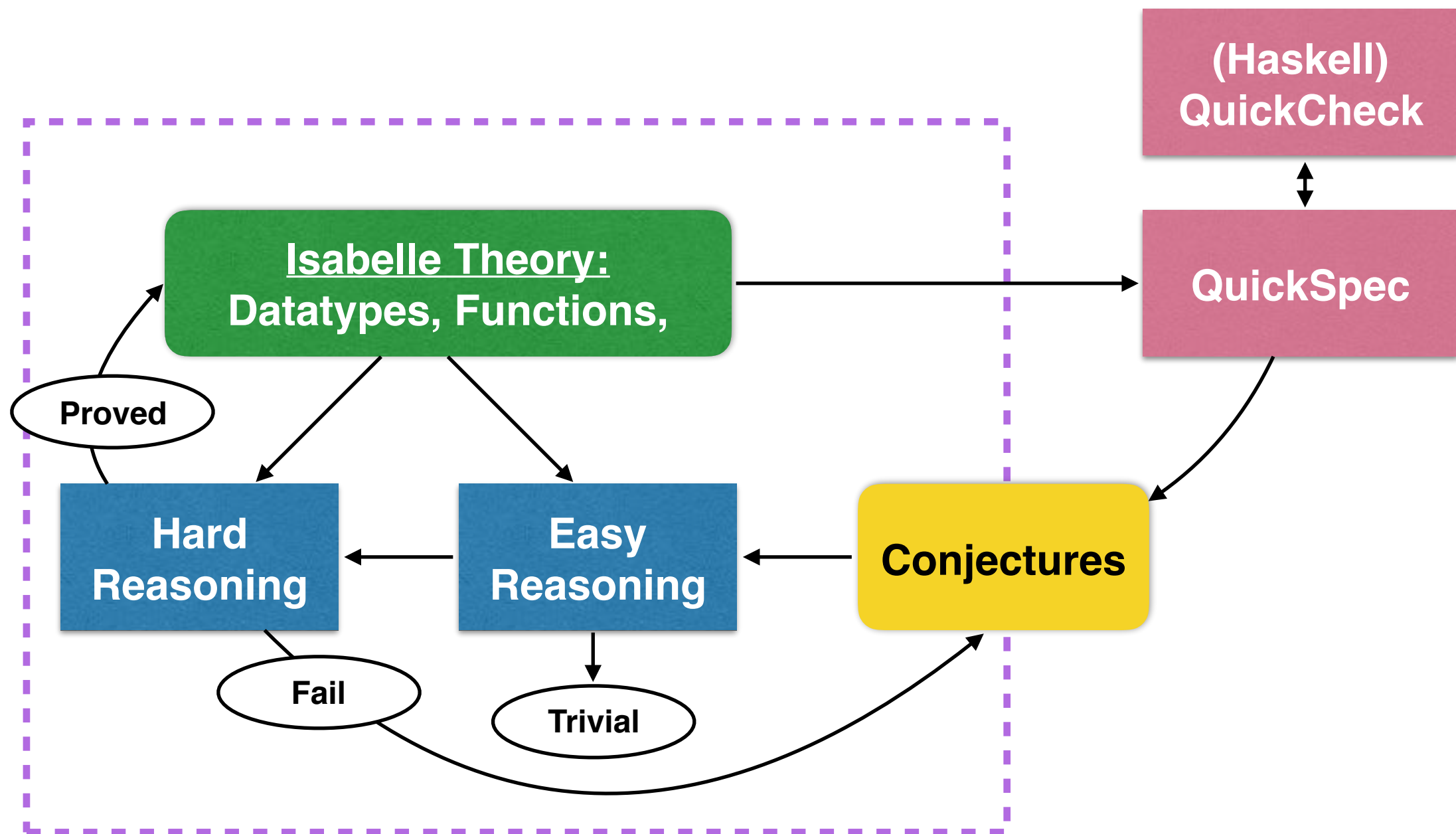
- What happens if we make a mistake in our code?
- **Demo:** unexpected properties...

```
fun len :: "'a list  $\Rightarrow$  nat"
  where
    "len [] = 0"
  | "len (x#xs) = Suc(len xs)"

fun rot :: "nat  $\Rightarrow$  'a list  $\Rightarrow$  'a list"
  where
    "rot 0 xs = xs"
  | "rot (Suc n) [] = []"
  | "rot (Suc n) (x#xs) = rot n xs@[x]"
```

Theorem discovery in Hipster

~~“Within ten years~~ a digital computer will discover and prove an ~~important new~~ mathematical theorem”



Try it!

- <https://github.com/moajohansson/IsaHipster>
- <http://hackage.haskell.org/package/quickspec>

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